Mines Paris PSL - Etilab

Dilution Rules and the Value of a Most-Favored-Nation Clause

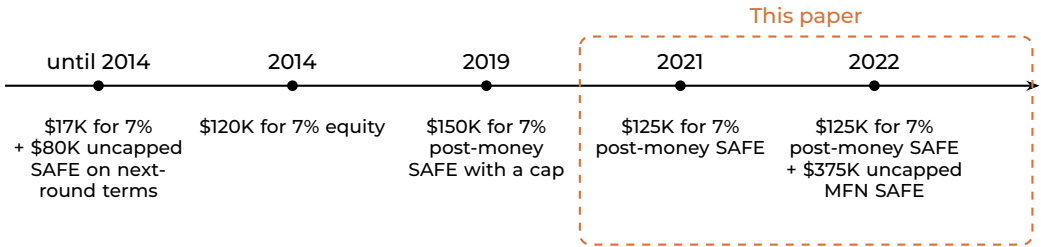
Stephen Dossou
Under the supervision of
Pierre Fleckinger and Philippe Mustar

October 21, 2026

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Chronology of Y Combinator deal terms



Since 2024, other accelerators have followed with their own MFN SAFEs.

Accelerators' MFN SAFEs

Since 2022, accelerators add an MFN SAFE to the deal they post for the whole batch.

	Fixed amount, fixed share	MFN SAFE	Launch
Y Combinator	$125K for 7%	$375K uncapped MFN safe	2022
Entrepreneur First	$125K for 8%	$125K uncapped MFN safe, US, optional	2024
Techstars	$20K for 5%	$200K uncapped MFN safe	2025
Antler Nordics	€100K for 8.5%	€100K uncapped MFN safe	2025
Antler UK	£125K for 8.5%	£85K convertible note with MFN	2025

The MFN SAFE takes the terms of the lowest cap SAFE the startup issues between the start of the batch and its next equity round.

Seed investment contracts

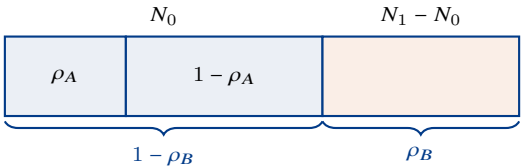
Two investors, A and B, invest in the firm one after the other.

- Investor i invests σ_i and buys the share ρ_i of the firm.
- The post-money valuation of its round is $V_i = \sigma_i / \rho_i$.
- With N_t shares and the price p_t of a share, $V = N_t\, p_t$.
- The pre-money valuation of investor B's round is $v_B = N_0\, p_1 = (1 - \rho_B)\, V_2$.

Investor B always runs an equity round. Investor A's contract sets the share it holds when this round converts it.

Seed contracts: two equity rounds

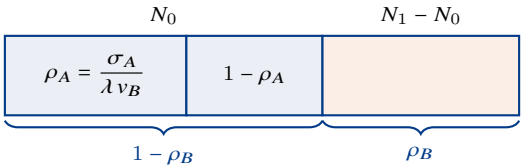
Each equity round dilutes the earlier shareholders in proportion to their holdings.



- Investor A: $\rho_A N_0 = \rho_A(1-\rho_B)N_1$ shares
- Investor B: $\rho_B N_1$ shares
- Entrepreneur: $(1-\rho_A)N_0 = (1-\rho_A)(1-\rho_B)N_1$ shares

Seed contracts: SAFE with a discount

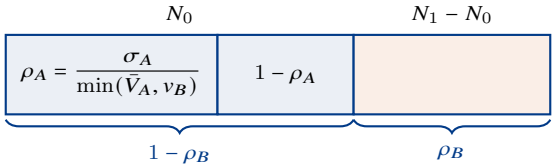
Investor A's SAFE converts at investor B's pre-money valuation, multiplied by the discount $\lambda \in (0, 1]$.



- Investor A: $\frac{\sigma_A}{\lambda v_B}(1 - \rho_B)N_1$ shares
- Investor B: $\rho_B N_1$ shares
- Entrepreneur: $\left(1 - \frac{\sigma_A}{\lambda v_B}\right)(1 - \rho_B)N_1$ shares

Seed contracts: capped SAFE

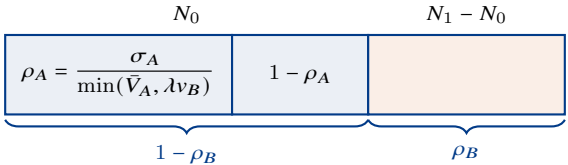
Investor A's SAFE converts at the lower of its cap and investor B's pre-money valuation.



- Investor A: $\frac{\sigma_A}{\min(\bar{V}_A, v_B)}(1 - \rho_B)N_1$ shares
- Investor B: $\rho_B N_1$ shares
- Entrepreneur: $\left(1 - \frac{\sigma_A}{\min(\bar{V}_A, v_B)}\right)(1 - \rho_B)N_1$ shares

Seed contracts: capped SAFE with a discount

Investor A's SAFE converts at the lower of its cap and the discounted pre-money valuation.

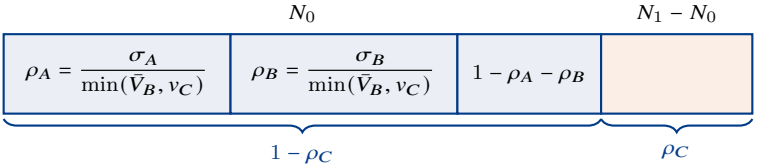


- Investor A: $\frac{\sigma_A}{\min(\bar{V}_A, \lambda v_B)}(1 - \rho_B)N_1$ shares
- Investor B: $\rho_B N_1$ shares
- Entrepreneur: $\left(1 - \frac{\sigma_A}{\min(\bar{V}_A, \lambda v_B)}\right)(1 - \rho_B)N_1$ shares

The uncapped MFN SAFE

Investor A's SAFE takes investor B's cap. Post-money SAFEs do not dilute one another before investor C's round.

Investor A holds an uncapped MFN SAFE, investor B a capped SAFE with no discount, investor C runs an equity round.



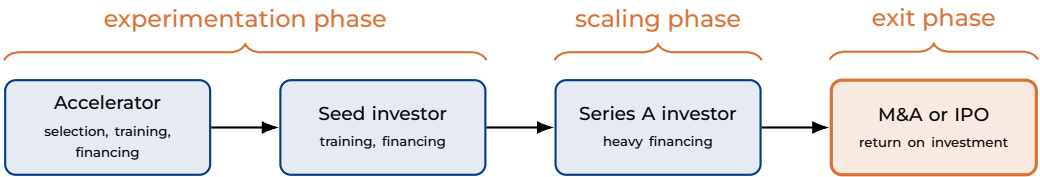
- Investor A: $\frac{\sigma_A}{\min(\bar{V}_B, v_C)}(1-\rho_C)N_1$ Investor B: $\frac{\sigma_B}{\min(\bar{V}_B, v_C)}(1-\rho_C)N_1$
- Investor C: $\rho_C N_1$
- Entrepreneur: $\left(1 - \frac{\sigma_A + \sigma_B}{\min(\bar{V}_B, v_C)}\right)(1-\rho_C)N_1$

Seed contracts: outside the scope

- Convertible notes work like SAFEs, with an interest component and a liquidation preference.
- Pre-money SAFEs can be diluted by other SAFEs before an equity round.

Both are outside the scope of this paper.

Venture financing sequence



A startup first pays to learn what its project is worth, then pays to scale it.

Kerr, Nanda and Rhodes-Kropf (2014), *Entrepreneurship as Experimentation*.

Literature Review

- Entrepreneurship as experimentation
  - Kerr, Nanda and Rhodes-Kropf (2014): a project's value is learned by investing in it.
  - Ewens, Nanda and Rhodes-Kropf (2018): cheaper experiments, smaller stakes in more startups.
  - Nanda and Rhodes-Kropf (2017): the risk that the next round fails shapes early financing.
- Securities in venture capital
  - Admati and Pfleiderer (1994): a fixed fraction removes the incentive to misprice later rounds.
  - Coyle and Green (2014): after 2005, capped convertible notes and then SAFEs finance the seed stage.
  - Hellmann and Thiele (2015): angels and VCs are friends and foes.
- Most-favored-customer clauses
  - Cooper (1986): the clause helps a seller keep prices high.
  - McAfee and Schwartz (1994): nondiscrimination clauses generally do not protect early signatories.

This paper

What is an MFN clause held by an accelerator worth to the startups that grant it?

Findings

- The value of the clause depends on the dilution rules of the seed contract.
- Without a seed round, the MFN SAFE replaces Series A money that diluted the accelerator.
- With a seed equity round, firms whose need is close to the seed claims lose their seed round.
- With a capped seed SAFE, only firms whose threshold is set by the discount can lose.
- All the losses come from the claim. The cash alone harms no firm.

The entrepreneur

The entrepreneur has no initial budget and bears two sunk costs.

A project

- A stochastic liquidity shock $l \sim G$, paid in the experimentation phase.
- A scaling cost I, known ex ante, paid in the scaling phase.
- The project is worth 1 if it is sold or goes public, 0 if it fails.
- Its success probability $\pi \sim F$ is revealed once the experimentation phase is done.
- Once the project is funded, the entrepreneur bears a private effort cost $c_e > 0$.

The financiers

experimentation phase

scaling phase

Accelerator

(σ_a, ρ_a)
Before l and π
Standard terms, taken as given

Seed investor

(σ_s, ρ_s)
After l, before π
Budget β
Monopolist

Series A investor

$(\sigma_A, \rho_A = \sigma_A/V)$
After π
Competitive

Assumptions

(A1) $\mathbb{E}(\pi) < I + c_e$. There is no scaling phase without the experimentation phase.

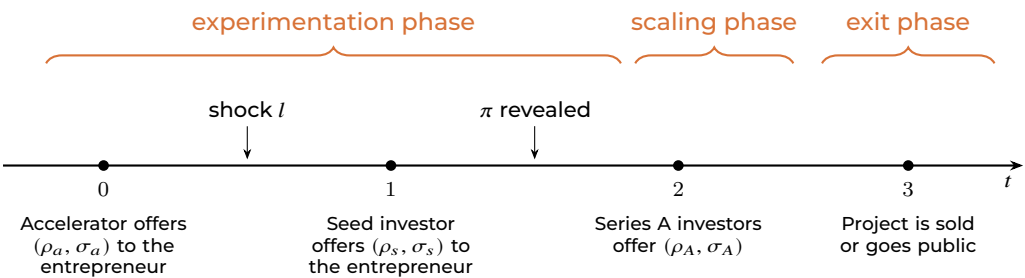
(A2) $\int_{I+c_e}^{1} (\pi - I - c_e)\, f(\pi)\, d\pi > \mathbb{E}(l)$. The experimentation phase is worth running.

(A3) f is positive, C^1 and log-concave.

(B1) $\sigma_a + \beta \leq I$. The accelerator and the seed investor cannot run the scaling phase.

A firm whose cash $\sigma_a + \sigma_s$ does not cover l is liquidated.

Timing


experimentation phase
scaling phase
exit phase
shock l
π revealed
0
1
2
3
t
Accelerator offers (ρ_a, σ_a) to the entrepreneur
Seed investor offers (ρ_s, σ_s) to the entrepreneur
Series A investors offer (ρ_A, σ_A)
Project is sold or goes public

Payoffs

For each project π that is fully financed:

$$u_e(\pi) = (1-\rho_a)(1-\rho_s)(1-\rho_A)\,\pi - c_e$$

$$\Pi_A(\pi) = \rho_A(\pi)\,\pi - \sigma_A = \sigma_A\left(\frac{\pi}{V(\pi)} - 1\right)$$

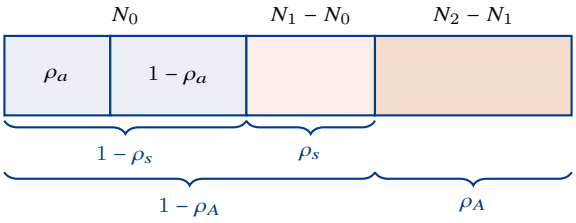
After the Series A round the seed investor holds $\rho_s\big(1-\rho_A(\pi)\big)$, so

$$\Pi_s = \int_{\pi_{\min}}^{1} \Big(\rho_s\big(1-\rho_A(\pi)\big)\,\pi - \sigma_s\Big) f(\pi)\,d\pi \;-\; \int_{0}^{\pi_{\min}} \sigma_s\, f(\pi)\,d\pi.$$

Benchmark: equity rounds only

The accelerator, the seed investor and the Series A investor each hold a share diluted by every later round.

Equivalently, the accelerator holds a post-money SAFE with a fixed share, and the seed and Series A rounds are equity rounds.



- Series A: $\rho_A N_2$ Seed: $\rho_s N_1 = \rho_s(1-\rho_A)N_2$
- Accelerator: $\rho_a N_0 = \rho_a(1-\rho_s)(1-\rho_A)N_2$
- Entrepreneur: $(1-\rho_a)N_0 = (1-\rho_a)(1-\rho_s)(1-\rho_A)N_2$

Benchmark: the Series A round

A larger Series A round, or a smaller share left to the founder, leaves fewer projects financed.

$$u_e(\pi) \geq 0 \qquad (IC_e)$$

$$\frac{\sigma_A}{V(\pi)}\pi - \sigma_A \geq 0 \quad \Longrightarrow \quad V^*(\pi) = \pi \qquad (IR_A)$$

$(IC_e) + (IR_A) \quad \Longrightarrow \quad$ the Series A finances $\pi \geq \sigma_A + \dfrac{c_e}{(1-\rho_a)(1-\rho_s)}$

Benchmark: the seed investor's problem

The share the seed investor takes decides which projects get a Series A.

$$\max_{\sigma_s, V_s} \Pi_s = \underbrace{\int_{\pi_{\min}}^{1} \left(\frac{\sigma_s}{V_s} (\pi - \sigma_A) - \sigma_s \right) f(\pi)\, d\pi}_{\text{these projects get a Series A}} - \underbrace{\int_{0}^{\pi_{\min}} \sigma_s\, f(\pi)\, d\pi}_{\text{these do not}}$$

The seed investor invests before π is revealed, so it cannot tell the two groups apart.

Lemma 1

The seed investor invests nothing or at least the need, $\sigma_s \geq l - \sigma_a$.

Benchmark: valuation and amount

Proposition 1 and Corollary 1

For each σ_s and σ_A,

$$V_s = \frac{\sigma_s}{1 - \dfrac{c_e}{(1-\rho_a)\big(\pi(\sigma_A) - \sigma_A\big)}},$$

where $\pi(\sigma_A)$ depends on f, c_e and ρ_a, and is strictly increasing in σ_A.

Proposition 2

A participating seed investor covers the need exactly, $\sigma_s^* = l - \sigma_a$. Cash beyond the need is lost on the projects that do not reach Series A.

Benchmark: three regimes

The liquidity shock sorts firms into no seed round, a seed round, and liquidation.

Lemma 2 and Proposition 3

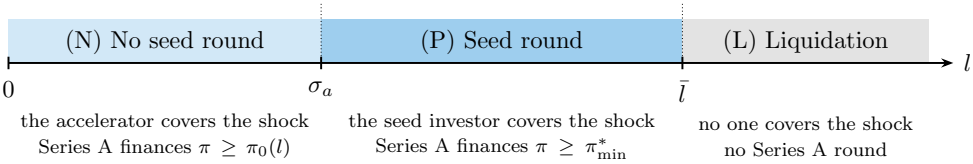
If $l \leq \sigma_a$, the entrepreneur turns down every seed offer. There is $\bar{l}$, with $\bar{l} - \sigma_a$ depending on f, I, c_e, ρ_a and β but not on σ_a, such that

(N) $l \leq \sigma_a$: the Series A finances $\pi \geq \pi_0(l) := I + l - \sigma_a + \dfrac{c_e}{1-\rho_a}$

(P) $\sigma_a < l \leq \bar{l}$: the Series A finances $\pi \geq \pi(\sigma_A^*)$

(L) $l > \bar{l}$: the firm is liquidated before Series A

Benchmark: three regimes



Benchmark: more accelerator cash

With more accelerator cash, more firms need no seed round and fewer are liquidated. Seed rounds can become more or less frequent.

Proposition 4

Raising the accelerator's amount from σ_a to σ_a' gives $\bar{l}' = \bar{l} + \sigma_a' - \sigma_a$, and

(i) raises the probability of no seed round from $G(\sigma_a)$ to $G(\sigma_a')$,

(ii) raises the probability of a seed round iff $G(\bar{l}') - G(\bar{l}) > G(\sigma_a') - G(\sigma_a)$,

(iii) lowers the probability of liquidation from $1 - G(\bar{l})$ to $1 - G(\bar{l}')$.

Benchmark: more accelerator cash

Proposition 5

More accelerator cash lowers no firm's probability of reaching Series A.

A firm that no longer needs a seed round gets a Series A for

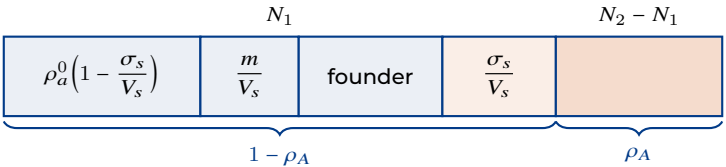
$$\pi \geq I + l - \sigma_a' + \frac{c_e}{1 - \rho_a},$$

which lies below $\pi(I)$, its lowest financed project with a seed round.

Seed equity round: the cap table

The MFN SAFE converts at the seed price. The seed round dilutes the accelerator's post-money SAFE, not the MFN SAFE.

Accelerator: post-money SAFE ρ_a^0 and uncapped MFN SAFE m. Seed and Series A: equity rounds.



- Series A: $\rho_A N_2$ Seed: $\frac{\sigma_s}{V_s}(1 - \rho_A)N_2$
- Accelerator: $\left(\rho_a^0\left(1 - \frac{\sigma_s}{V_s}\right) + \frac{m}{V_s}\right)(1 - \rho_A)N_2$
- Entrepreneur: $\left((1 - \rho_a^0)\left(1 - \frac{\sigma_s}{V_s}\right) - \frac{m}{V_s}\right)(1 - \rho_A)N_2$

Seed equity round: the MFN SAFE

Without a seed round, the MFN SAFE converts at the Series A price and is worth m on every financed project.

$$\pi \;\geq\; \pi_0^m(l) := I + l - \sigma_a + \frac{c_e + m}{1 - \rho_a^0}$$

The Series A round it replaces diluted the accelerator's post-money SAFE. The MFN SAFE does not.

Seed equity round: valuation and amount

Proposition 6

The seed investor proposes

$$V_s^* = \frac{\big((1-\rho_a^0)\sigma_s + m\big)\big(\pi(\sigma_A) - \sigma_A\big)}{(1-\rho_a^0)\big(\pi(\sigma_A) - \sigma_A\big) - c_e},$$

and the Series A finances the projects above $\pi(\sigma_A)$, as in the benchmark.

Proposition 7

If $m > 0$ and the SAFE block is worth more than m at exact funding, the seed investor invests beyond small needs.

Seed equity round: the reform

With a seed equity round, the reform liquidates the firms whose need comes close to the value of the seed claims.

The reform: $\sigma_a = \sigma_a^0 + m$, ρ_a^0 unchanged. Write $x := l - \sigma_a^0$ and $\pi_0 := I + x + c_e/(1-\rho_a^0)$.

Proposition 9

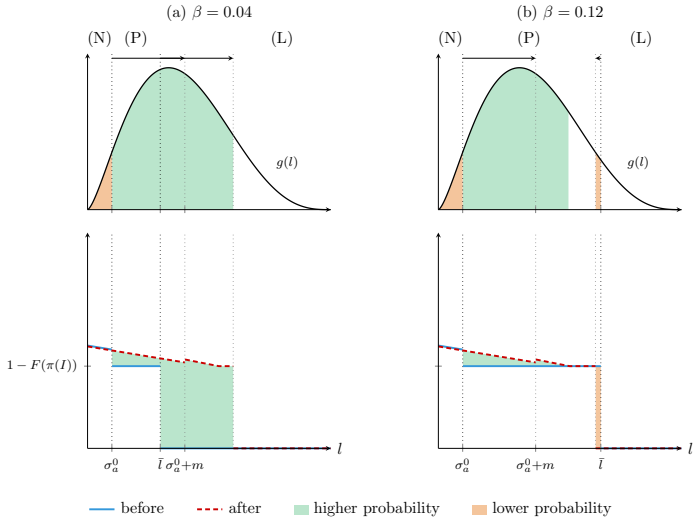
(i) $x \leq 0$: the probability of reaching Series A falls if $\pi_0 < 1$.

(ii) $0 < x \leq m$: it falls iff the firm had a seed round and $x > \pi(I) - I - \dfrac{c_e + \rho_a^0 m}{1-\rho_a^0}$.

(iii) $x > m$: the firms with $W(I) - \dfrac{\rho_a^0 m}{1-\rho_a^0} < x \leq \min\{\beta, W(I)\}$ lose their seed round.

$W(I)$ is the value of a benchmark seed investor's claims at exact funding.

Seed equity round: the reform

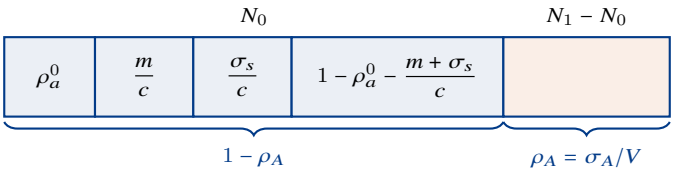


$\pi \sim U[0,1]$, $I = 0.5$, $c_e = 0.005$, $\rho_a^0 = 0.07$, $\sigma_a^0 = 0.02$, $m = 0.06$.

Capped seed SAFE: the cap table

The SAFE block $m + \sigma_s$ converts at $c := \min(V_s, \lambda v)$, with $v = V - \sigma_A$ the Series A pre-money valuation.

Accelerator: post-money SAFE ρ_a^0 and uncapped MFN SAFE m. Seed: capped SAFE with a discount. Series A: equity round.



- Series A: $\rho_A N_1$ Seed: $\frac{\sigma_s}{c}(1 - \rho_A)N_1$
- Accelerator: $\left(\rho_a^0 + \frac{m}{c}\right)(1 - \rho_A)N_1$
- Entrepreneur: $\left(1 - \rho_a^0 - \frac{m + \sigma_s}{c}\right)(1 - \rho_A)N_1$

Capped seed SAFE: conversion

The SAFE block $m + \sigma_s$ converts at the lower of the cap and the discounted Series A pre-money valuation.

Case 1: converts at the cap

$$\pi \geq \sigma_A + \frac{c_e}{1 - \rho_a^0 - (m + \sigma_s)/V_s}$$

Case 2: converts at the discount

$$\pi \geq \pi^d_{\min} := \sigma_A + \frac{c_e + \frac{m+\sigma_s}{\lambda}}{1 - \rho_a^0}$$

Post-money SAFEs do not dilute one another. In Case 2 the block's claims are a fixed amount.

Capped seed SAFE: the cap

Proposition 10

For each σ_s, the seed investor sets the threshold $x^* := \max\{\pi(\sigma_A), \pi^d_{\min}\}$ with the cap

$$V^*_s = \frac{(m + \sigma_s)(x^* - \sigma_A)}{(1 - \rho^0_a)(x^* - \sigma_A) - c_e}.$$

Every financed project converts at the cap.

Case 1, $\pi(\sigma_A) \geq \pi^d_{\min}$: the threshold of a seed equity round. Case 2: the lowest threshold the cap can reach.

In Case 1 the cap leaves the founder the same share as a seed equity round.

Capped seed SAFE: the amount

The seed investor shares the SAFE block with the MFN SAFE, and a dollar it adds raises its fraction $\sigma_s/(m+\sigma_s)$.

Proposition 11

(i) If $m = 0$, the seed investor covers the need exactly.

(ii) If $m > 0$, Case 1 holds at small needs, and the SAFE block is worth more than m at exact funding, the seed investor invests beyond small needs.

Proposition 12

Three regimes as in the benchmark, with $\pi \geq \pi_0^m(l)$ in (N) and $\pi \geq \max\{\pi(\sigma_A^*), \pi_{\min}^d\}$ in (P).

Capped seed SAFE: the reform

The reform changes the amount placed in SAFEs, not the map from this amount to the projects financed.

S is the amount in SAFEs beyond the former cash, σ_s before the reform and $m + \sigma_s$ after it.

Threshold of a firm with a seed round

$$T(S) = \max\left\{\pi\big(I + x - S\big),\ \pi_0 + \left(\frac{1}{\lambda(1-\rho_a^0)} - 1\right)S\right\}$$

- Before the reform the seed investor covers the need, $S = x$.
- Case 2 holds before the reform iff $x > \lambda\big((1-\rho_a^0)(\pi(I) - I) - c_e\big)$.
- If $x > m$, a firm with a seed round keeps it, with $S \geq x$.

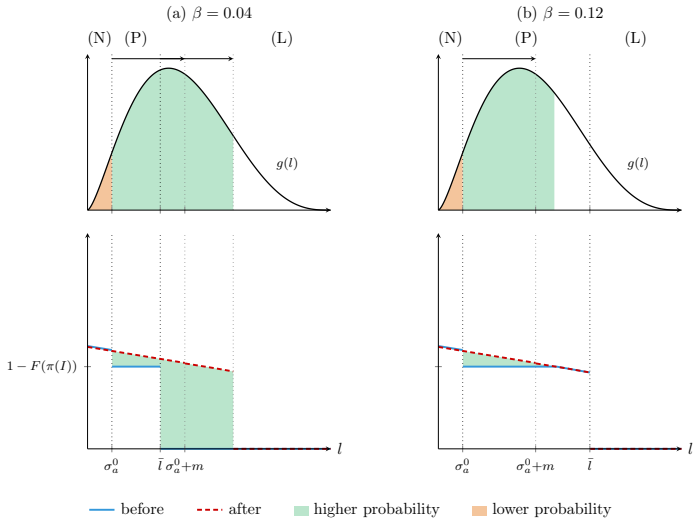
Capped seed SAFE: the reform

Firms that need no seed round lose, and so do firms in Case 2 that receive more in SAFEs.

Shock	Before the reform	Probability of reaching Series A
(i) $x \leq 0$	No seed round	$-$
(ii) $0 < x \leq m$	Liquidated	$+$
	Seed round	$+$ or $-$
(iii) $x > m$	Liquidated	0 or $+$
	Seed round, Case 1 after the reform	$+$
	Seed round, Case 2 before the reform	$-$
	Seed round, Case 1 before, Case 2 after	$+$ or $-$

Proposition 13. The signs are weak.

Capped seed SAFE: the reform



$\lambda = 1$. Same parameters as for the seed equity round.

Capped seed SAFE: equity round or SAFE

Proposition 15

At every amount, a seed equity round gives the seed investor at least as much as a capped SAFE.

The equity round dilutes the accelerator, the post-money SAFE does not. If $m = 0$, every firm's probability of reaching Series A is at least as high with the equity round.

Capped seed SAFE: cash and claim

All the losers of the reform come from the claim, not from the cash.

Proposition 16

Under both seed contracts:

(i) the cash m alone lowers no firm's probability of reaching Series A;

(ii) the claim m raises by $m/(1-\rho_a^0)$ the lowest financed project of every firm without a seed round.

Design of the capped SAFE

Proposition 17

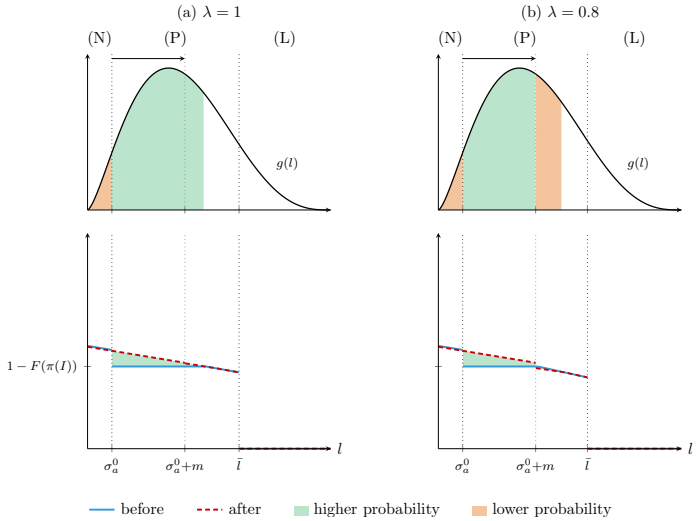
The seed investor's profit weakly increases with λ. It proposes a cap and no discount, $\lambda = 1$.

Proposition 18

Financing the same projects, a cap gives the seed investor more than a discount without a cap, which works as an uncapped convertible note.

With a discount set by the market, for instance $\lambda = 0.8$, Case 2 holds at more amounts, and a firm in Case 2 loses more projects when the reform raises its amount.

Design of the capped SAFE: a market discount



$\beta = 0.12$. Same parameters as for the seed equity round.

Takeaways

Main result

The value of the MFN clause depends on the dilution rules of the seed contract.

- Firms that need no seed round lose, because the MFN SAFE does not dilute the accelerator.
- With a seed equity round, the reform can liquidate firms that had a seed round.
- With a capped seed SAFE, firms in Case 2 that receive more in SAFEs lose.
- All the losses come from the claim. The cash alone harms no firm.
- Seed investors prefer a capped SAFE with no discount.

Thank you

Dilution Rules and the Value of a Most-Favored-Nation Clause

Stephen Dossou

Mines Paris PSL – Etilab

stephen.dossou@minesparis.psl.eu

stephendossou@gmail.com

eti|lab

MINES PARIS | PSL